BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: PR – XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR – XII (01) (23)

Note: Four choices are given against each question; fill the circle of the correct answer only. Use a black or blue ball point / marker to fill the circle. Filling more than one circle for a question will be counted as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Grignard's reagent contains ____ metal.

(A) Fe (B) Ca

(C) Mg (D) Mn

Q2. The bond angles of H-C-H and H-C-Cl bonds are ____.

(A) Less than 109.5° (B) Greater than 109.5°

(C) Equal to 109.5° (D) Equal to 120°

Q3. 1,2,4,5-tetramethylbenzene is the IUPAC name of ____.

(A) Xylene (B) Toluene

(C) Durene (D) Mesitylene

Q4. Optica activity of a compound is measured by an instrument called ____.

(A) Photometer (B) Refractometer

(C) Polarimeter (D) Calorimeter

Q5. All of the given are alkanes except ____.

(A) C_3H_8 (B) C_4H_{10}

(C) C_6H_{12} (D) C_7H_{16}

Q6. Oxa is the functional group of ____.

(A) Ester
(B) Aldehyde
(C) Ether
(D) Ketone

Q7. Which of the given is not organic in nature?

(A) $NH_3CO\text{-}NH_2$
(B) NH_4CNO
(C) $CH_3\text{-}OH$
(D) CH_3Cl

Q8. The most stable oxidation state of manganese in acidic or neutral solution is ____.

(A) +2
(B) +4
(C) +6
(D) +7

Q9. Al_2O_3 is ____.

(A) Acidic oxide
(B) Basic oxide
(C) Neutral oxide
(D) Amhoteric oxide

Q10. Which of the given is NMR inactive nucleus?

(A) ^{31}P
(B) ^{1}H
(C) ^{15}N
(D) ^{16}O

Q11. In combustion analysis CO_2 is absorbed by ____.

(A) $Mg(ClO_4)_2$
(B) KOH
(C) CuO
(D) $CuSO_4$

Q12. The height of the stratosphere is from ____.

(A) 1 to 10 km
(B) 10 to 50 km
(C) 50 to 85 km
(D) 85 to 387 km

Q13. Total number of chlorine atoms in DDT are ____.

(A) 3
(B) 4
(C) 5
(D) 6

Q14. Proteins are polymers of ____.

(A) Glucose
(B) Disulphide bridges
(C) Peptide bonds
(D) Amino acids

Q15. All are the derivatives of carboxylic acid except ____.

(A) Acid Halides
(B) Nitrile
(C) Alkyl Halides
(D) Ester

Q16. Gem-Diol is the product of acid/base catalyzed reaction of water with ____.

(A) Aldehyde/ketone
(B) Carboxylic acid
(C) Alcohol
(D) Ether

Q17. In Williamson's synthesis ____ pair of reactants, react with each other to form ether.

(A) R-OH and Na
(B) R-OH and R-Mg-X
(C) R-OH and R-O-Na^+
(D) R-OH and R-OH

Q18. Lucas reagent is ____.

(A) R-OH and HCl
(B) HCl and $ZnCl_2$
(C) R-OH and $ZnCl_2$
(D) None of these

SECTION-B

Time Allowed: 2 Hours 40 Minutes

Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FOUR marks.

Q1. Write the chemical reactions to show that Al_2O_3 is amphoteric in nature?

Q2. Define electron affinity. Discuss its variation in Halogens?

Q3. Why compounds of Cu+ are diamagnetic while Cu2+ are paramagnetic?

Q4. What is homologous series? Give the characteristics of homologous series?

Q5. Enlist four physical properties of cycloalkanes?

Q6. The reaction of H-Br with H-C≡C-H gives 1,1-dibromoethane while with CH_3-C≡C-H, it gives 2,2-dibropropane. Give reason.

Q7. Write the IUPAC names of the following. (a) $C(CH_3)_4$ (b) CH_3-OH over CH_3-CH-CH-CH_3 (two-level as printed) (c) $(CH_3)_3$C-CH_2-C(=O)-H (d) HOOC-CH_2-COOH

Q8. Why secondary amines are more basic than tertiary amines in aqueous solution?

Q9. Steric hindrance makes the ketones less reactive than aldehydes. Comment over it.

Q10. Discuss the reactions of phenol with; (a) Bromine (b) Sodium metal

Q11. What is the effect of substituents on the acidity of carboxylic acids, if attached with carboxylic acid molecule?

Q12. State the medical problems that may relate to deficiencies of calcium and phosphorus?

Q13. Give the applications of infrared spectroscopy.

SECTION-C

Time Allowed: 2 Hours 40 Minutes

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) What is acid rain? Discuss its effects on environment and living organisms?

(b) Describe four fractions obtained by fractional distillation of petroleum?

Q4.

(a) What are elimination reactions? Discuss E1 mechanism in detail.

(b) How are aldehydes and ketones reduce to; (i) Alkanes by Wolf-Kishner Reduction (ii) Alcohols by Catalytic Method

Q5.

(a) Define stereoisomerism and explain its types?

(b) How carbon and hydrogen can be detected in organic compounds?

Q6.

(a) Draw the structural formula of the following compounds; (i) Naphthalene (ii) Acetic Anhydride (ii) Adipic Acid (iv) 2,4,6-Trinitrotoluene (v) 3,4-dimethylhexan-2-ol

(b) Differentiate between partial and total synthesis.